

Question	Marking Guidance	Mark	Comments
01.1	1. Draw line/origin on (chromatography) paper (using ruler and pencil); 2. (Use pipette/tubing/dropper to) add chlorophyll/solution to origin/line; 3. Add solvent/ A below line/origin; 4. Remove/stop (from glassware) before solvent reaches end (of chromatography paper) OR Mark (position) where solvent reaches/front;	4 (4 x AO1)	Accept all marks on a labelled diagram. Allow filter paper for chromatography paper. 3. Accept 'so solvent is not above line' for 'solvent below line'. 4. Accept remove/stop (immediately) when solvent reaches end/top. 4. Accept ensure solvent does not reach top/end. 4. Ignore Rf values.

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01.2	For 2 marks = two pigments have same solubility in (solvent) A but different solubility in (solvent) B OR For 2 marks = one pigment is soluble in (solvent) B but not in (solvent) A OR For 2 marks = pigments have different solubilities in the solvents/ A and B OR For 2 marks = one pigment is only soluble in (solvent) B OR For 2 marks = one pigment is insoluble (only) in (solvent) A ;; For 1 mark = response as for 2 marks but refers to solution rather than solvent;	2 (2 x AO3)	Accept 'dissolves' for solubility. Accept correct reference to 'affinity' for solubility e.g., 'greater affinity' (to solvent) indicates more soluble. Accept correct reference to 'polarity' for solubility' e.g. pigments have different polarities. Accept 'Rf value' for solubility. Ignore 'solvent stronger'.
